TECHFEST PROTOTYPE • IoT WATER MANAGEMENT

JALRAKSHAK

Smart IoT System to Monitor Water Distribution, Detect Leakages & Prevent Theft

Real-time flow monitoring, automated leak detection, and instant alerts to protect every drop of water.

CATEGORY
IoT Smart Water Management

PURPOSE
TechFest Prototype Demo

CENTRE
DigiSkill Khar Road

Save Water. Save Life. Smart Monitoring. Better Future.

THE PROBLEM

Water Loss Is a Growing Crisis

Silent Leakages

Pipeline leaks in distribution networks often go undetected for long periods, wasting huge volumes of water.

Water Theft

Unauthorized tapping and illegal connections divert water away from intended supply, causing shortages.

Delayed Detection

Without sensors, leaks and theft are usually discovered only after major loss or complaints.

Irregular Supply

Undetected loss and theft disrupt planning, leading to inconsistent and unfair water distribution.

JalRakshak ensures efficient monitoring and management of water distribution — detecting leaks and theft in real time.

CAPABILITIES

Key Features of JalRakshak

Real-Time Flow Monitoring

Continuously tracks water flow at both inlet and outlet points.

Leakage Detection

Identifies leaks by comparing inlet and outlet flow readings.

Water Theft Detection

Flags abnormal flow patterns that indicate unauthorized tapping.

Automatic Valve Control

Shuts off the control valve automatically when danger is detected.

Instant Alerts

Sends immediate LED, buzzer, and notification-based alerts.

IoT Remote Monitoring

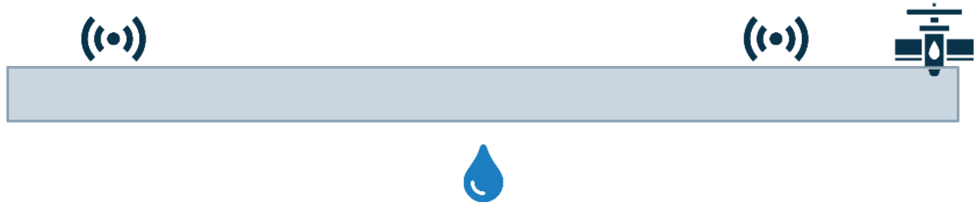
Enables dashboard and mobile-based monitoring from anywhere.

SYSTEM IN ACTION

Two Scenarios: Safe vs. Danger

Twin flow sensors at the inlet and outlet continuously compare readings to detect leakage or theft.

SCENE 1: NORMAL FLOW

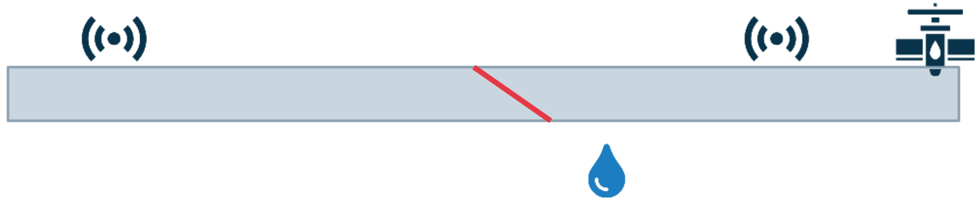


STATUS: SAFE

- ✔ Flow Normal, No Leakage
- ✔ No Theft Detected
- ✔ Green LED ON
- ✖ Buzzer OFF

Control Valve: OPEN

SCENE 2: LEAKAGE / THEFT



STATUS: DANGER

- ✖ Abnormal Flow Detected
- ✖ Possible Leakage / Theft
- ✖ Red LED ON
- ✔ Buzzer ON — Alert Sent

Control Valve: CLOSED

HOW THE SYSTEM IS STRUCTURED

System Architecture

1. FIELD LAYER

Sensors & Actuators

- Flow Sensor (Inlet)
- Flow Sensor (Outlet)
- Control Valve, LEDs, Buzzer

2. COMMUNICATION

IoT Connectivity

- ESP8266 NodeMCU
- Sends data via Wi-Fi
- to Cloud Server

3. CLOUD LAYER

Data Processing

- Receives & stores data
- Analyzes flow difference
- Triggers alerts

4. APPLICATION

Dashboard & Alerts

- Web dashboard
- Mobile notifications
- Email / SMS alerts

5. USERS

Stakeholders

- Water Supply Authorities
- Municipal Corporations
- Maintenance Team, Citizens

HARDWARE

Components Used

01

ESP8266 NodeMCU

Main microcontroller & Wi-Fi module

02

YF-S201 Flow Sensor (x2)

Measures inlet & outlet water flow

03

Motorized / Solenoid Valve

Controls water flow automatically

04

Green LED

Indicates safe condition

05

Red LED

Indicates danger condition

06

Buzzer

Sounds an audible alert

07

Power Supply (5V)

Powers the complete circuit

08

Connecting Wires

Circuit interconnections

PROCESS FLOW

How It Works

1

Flow Sensors

measure water flow at inlet and outlet

→

2

ESP8266

reads data and sends it to the cloud

→

3

Cloud Analysis

detects abnormal flow difference

→

4

Alert Triggered

LED, buzzer & notifications activate

→

5

Authorities

take action based on the alert

Status Indication (LED & Buzzer)

CONDITION	GREEN LED	RED LED	BUZZER
Safe (Normal Flow)	ON	OFF	OFF
Danger (Leakage / Theft)	OFF	ON	ON

Why It Matters

Simple, unmistakable LED and buzzer cues let field staff and citizens instantly recognize a leak or theft event — no technical training required.

Project Guides & Participants

Project Guides

MK Mr. Karan

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Centre: DigiSkill Khar Road

Category: IoT Based Smart Water Management System

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Thank You

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DigiSkill Khar Road | Guides: Mr. Karan, Ms. Priyanka